

KAL PENEV

Senior Software Engineer

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EU and UK citizenship

<https://penev.me>

ABOUT

- Passionate about learning new skills, writing nice code, and working with great people.
- Strong interest in designing highly performant, scalable, and architecturally solid software.
- Natural problem solver with an affinity for complex technical challenges that require focus and independence.
- Meticulous about algorithm efficiency, memory management, optimization, semantics and safety.
- Drawn to powerful system programming languages such as C++ and Rust.

PROJECTS

[cpp-wasm-template](#): **C++23** repository template with static analysis, testing, CI and automatic web page deployment through compilation to **wasm**.

(wip) [matching-engine](#): **C++23** matching engine core implementation. Exploration of design patterns and optimizations in modern HFT systems - lock-free data structures, memory contention and cache locality, kernel bypass, concurrency patterns, etc.

[wgpu-hacking](#): Implementation of a white paper on rendering grass with curves using instanced rendering in **Rust** and **wgpu**.

[raytracing-weekend](#): Implementation of the book on path tracing in **Rust**.

[gpu-gems-scan](#): memory-coalesced parallel sum algorithm in **CUDA** with optimized bank conflicts for the Nvidia GPU architecture.

[eudaimonia](#): 3D Game Engine written in **C++** and **OpenGL** with a focus on performance and zero overhead abstraction. Data-driven and cache-coherent architecture. Physically based rendering. (uni)

[daydreaming](#): A small game created in **C++** with **SDL2**. (uni)

[cpu-3d-renderer](#): CPU-based rasterizer capable of rendering simple 3D shapes. (uni)

[hand posture recognition](#): Convolutional NN (YOLO v4) that recognizes hand poses in varying degrees of self-occlusion and lighting in real time using a phone camera stream. Trained and deployed with **Python** and **TensorFlow**. (uni)

EDUCATION

BSc Games Software Engineering
First Class Honours
Bournemouth University
09/2018 - 06/2021

EXPERIENCE

Arm

Cambridge, UK

Senior Software Engineer 10/2025 - Present

Software Engineer 11/2024 - 10/2025

- Architected and implemented a **C++23** application for image quality validation with a focus on determinism, compile-time safety, performance and extendability using functional programming patterns.
- Lead development of a high performance **C++** SDK with a **C** API to deliver rendering and neural algorithms implemented in **Vulkan**.
- Design and implement highly parallel algorithms optimized for Arm GPUs.
- Integration and optimization of hardware-accelerated ML techniques for image upscaling and framerate upsampling using **Vulkan** and **C++**.
- Reverse-engineering and optimizing various **C++** codebases (game engines, frameworks) with respect to rendering performance.
- Profiling and debugging of CPU and GPU workloads on Linux, Windows and Android.
- Writing compute GPU shaders/kernels optimized for throughput and memory bandwidth on the Mali GPU architecture in **GLSL** and **HLSL**.
- Delivered a talk on Neural Algorithms for mobile GPUs at [Vulkanised 2026](#).

Hawk-Eye Innovations (Sony Sports)

London, UK

Software Engineer 04/2023 - 11/2024

Junior Software Engineer 09/2021 - 04/2023

- Maintained and improved a Virtual Production system for electronic sports officiating in **C++20** with graphics rendered in **Unreal Engine**, orchestrated via a **Qt6** GUI app, and a custom **TCP/IP** back-end architecture for communication with other parts of the software pipeline, such as video and tracking data ingestion and broadcast synchronization.
- Took the lead in designing and implementing a new architecture for the main rendering product starting from the ground up. It was optimised for the business's needs (real time data flow, broadcast synchronization, determinism, playback rate manipulation) using functional principles and template metaprogramming in **C++20**. This included crafting **technical design documents** and diagrams, introducing **architectural decision records**, presenting the idea to stakeholders, and writing all of the code for the initial MVP.
- Ownership over design, implementation and maintenance of multiple features related to simulation logic, multi-threading, rendering, networking and operations.
- Personally led a transition to **Test-Driven Development** in the team with **GoogleTest** in our **C++** infrastructure and Unreal Test Automation for our rendering product. This was crucial to uphold sports integrity.
- Developed a real time fully **data-driven** skeletal character animation system using a stream of tracking data in proprietary format.
- Peer reviewed code, on-boarded, and mentored teammates. Experienced cross-team dependency coordination and deliveries with deadlines for the largest sports events in the world, such as FIFA World Cup, Wimbledon, NBA, UEFA Champions League and many others.

VISTA AR

Bournemouth, UK

Junior Developer 02/2020 - 08/2020

A digital technology combining the real and virtual worlds in order to discover the invisible history of a site.

- Development of AR apps in **Unity** and **C#**.
- Profiling and performance optimization for mobile in high poly scenes.
- Shader programming with **HLSL** and Shader Graph.
- Performance analysis with **Python**.

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